

EXPERIMENT: 8

STEERING ANGLE VARIATION IN MIMO BEAMFORMING USING MATLAB

OBJECTIVE – 1: Analyze Steering Angle (Degrees)

MATLAB Program

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

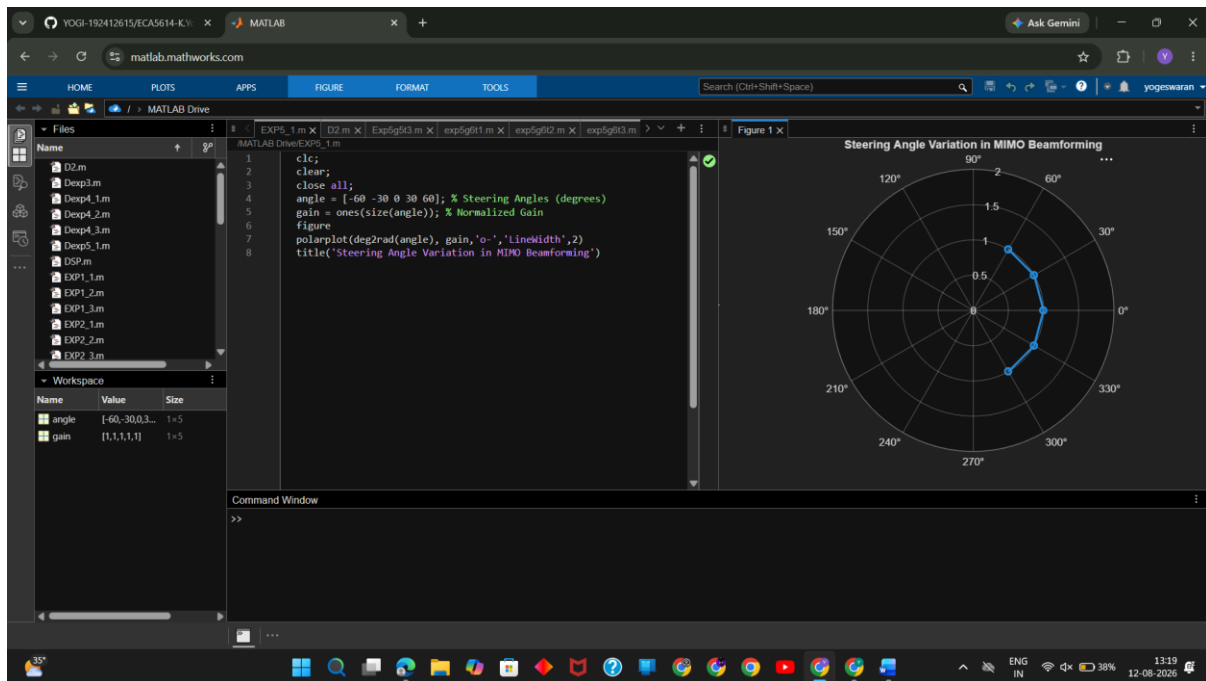
gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

OUTPUT



Objective – 2 Compare Array Gain (dB)

Matlab Code :

```
clc;
```

```

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure

bar(angle, gain)

grid on

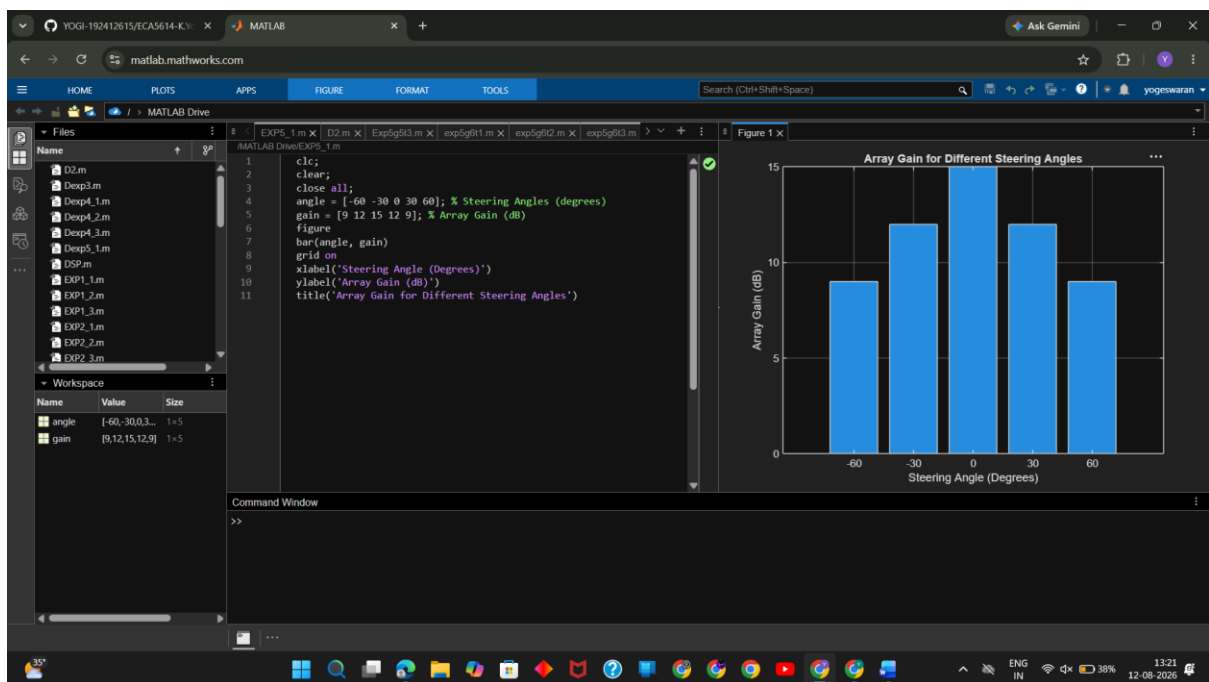
xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')

```

OUTPUT



Objective – 3 Analyze Main Lobe Direction (Degrees)

Matlab Code

```

clc;

clear;

close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)

actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)

error = abs(desired - actual); % Beam Direction Error

figure

```

```
plot(desired, error, 'o-', 'LineWidth', 2)
```

```
grid on
```

```
xlabel('Desired Steering Angle (Degrees)')
```

```
ylabel('Beam Direction Error (Degrees)')
```

```
title('Beam Direction Error for Different Steering Angles')
```

OUTPUT

